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(54) SEMICONDUCTOR DEVICE WITH SUPPORTER AGAINST WHICH BONDING WIRE IS DISPOSED AND METHOD FOR PREPARING THE SAME

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***H01L 23/00*** (2006.01)

(52) **U.S. Cl.**
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See application file for complete search history.

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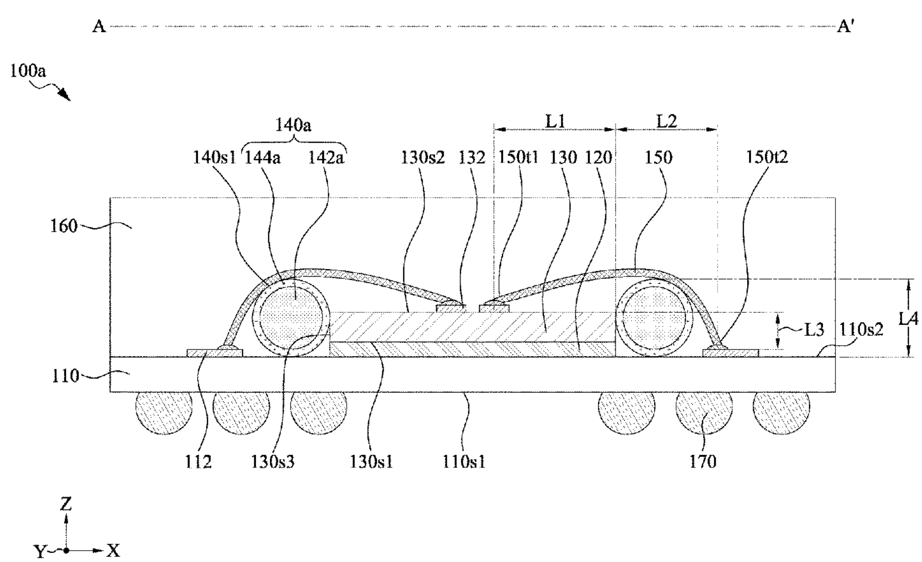
(57) **ABSTRACT**

A semiconductor device and method for manufacturing the same are provided. The semiconductor device includes a substrate, an electronic component, a bonding wire, and a supporter. The electronic component is disposed on the substrate. The bonding wire includes a first terminal connected to the electronic component and a second terminal connected to the substrate. The bonding wire is disposed against the supporter.

**19 Claims, 14 Drawing Sheets**



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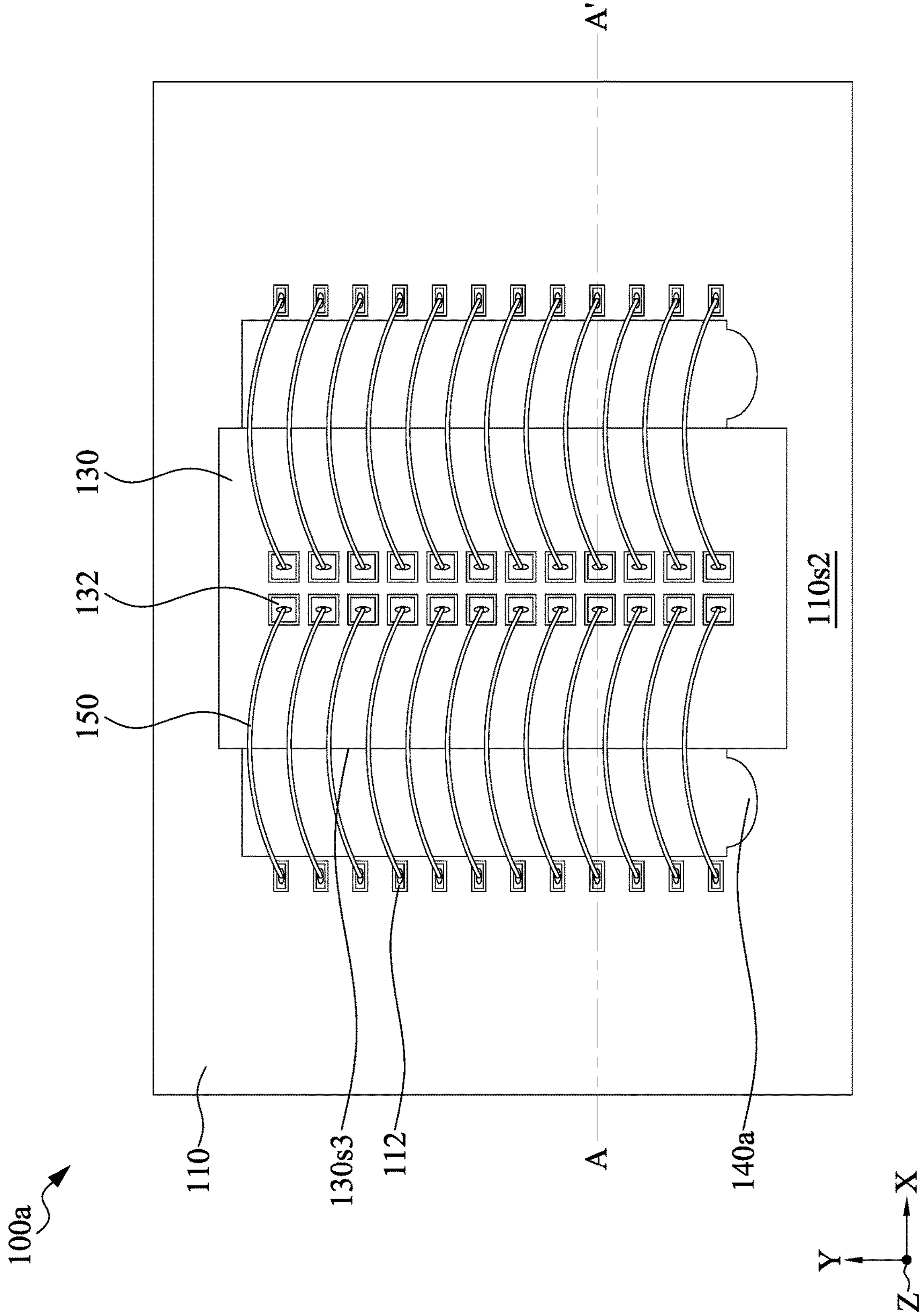



FIG. 1A

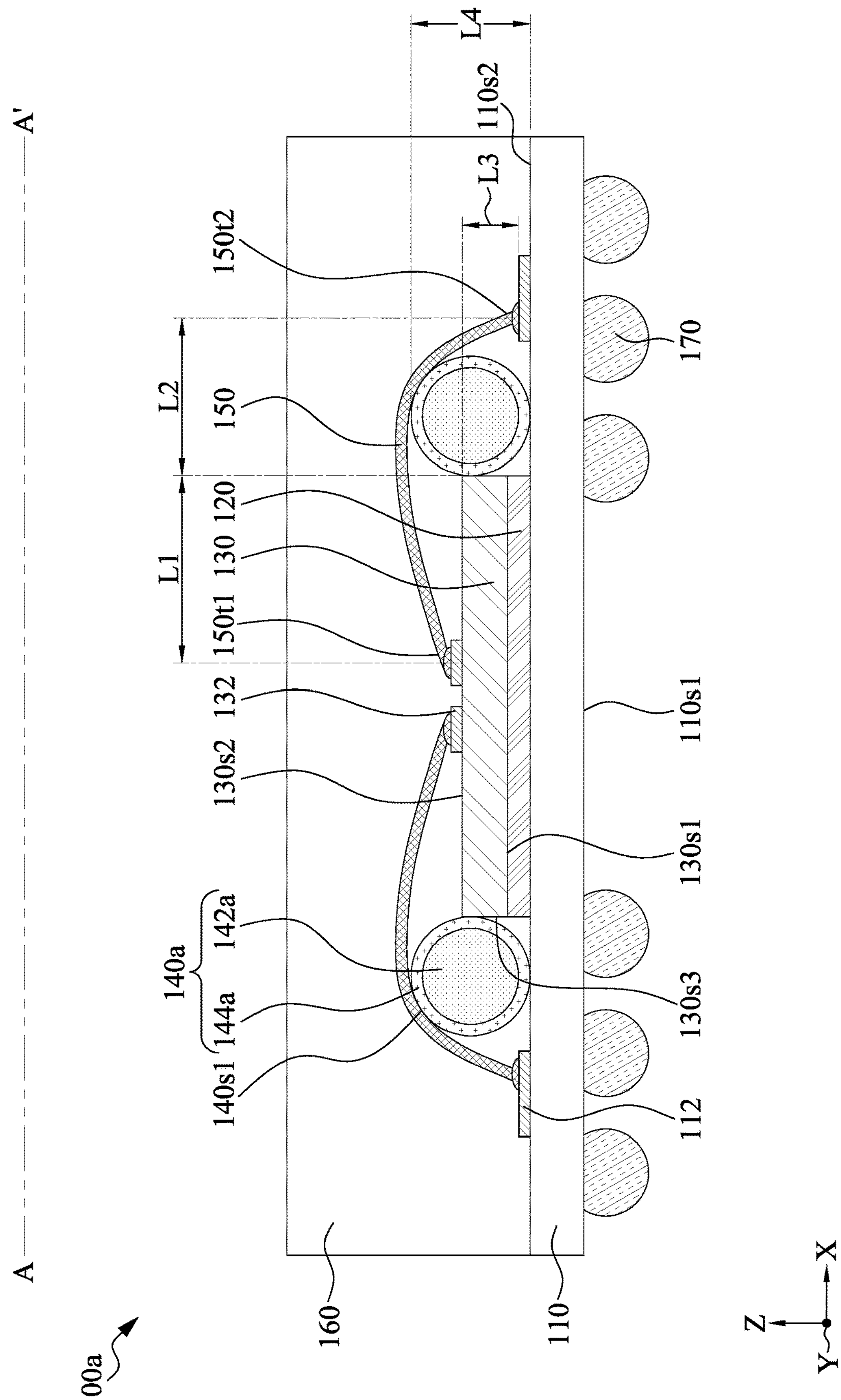
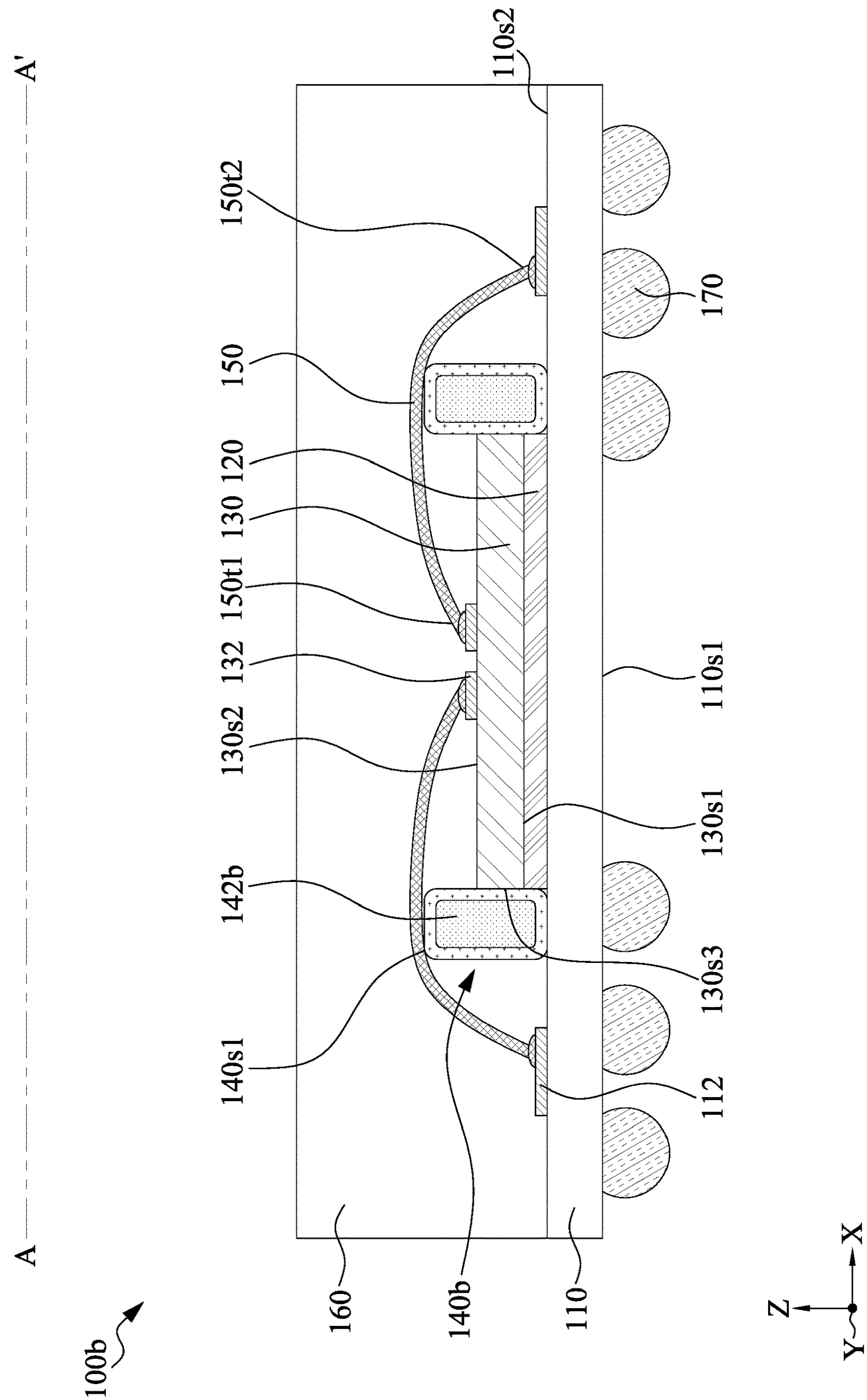

A
A'
L4
110s2
L3
150t2
170
L2
150
120
L1
130
150t1
132
110s1
130s2
130s1
142a
140a
144a
130s3
140s1
112
160
110
100a
X
Z
Y

FIG. 1B


A'
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150t2
170
150
120
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150t1
132
110s1
130s2
130s1
142b
130s3
140s1
112
A
160
140b
110
X
Z
Y
100b

FIG. 2

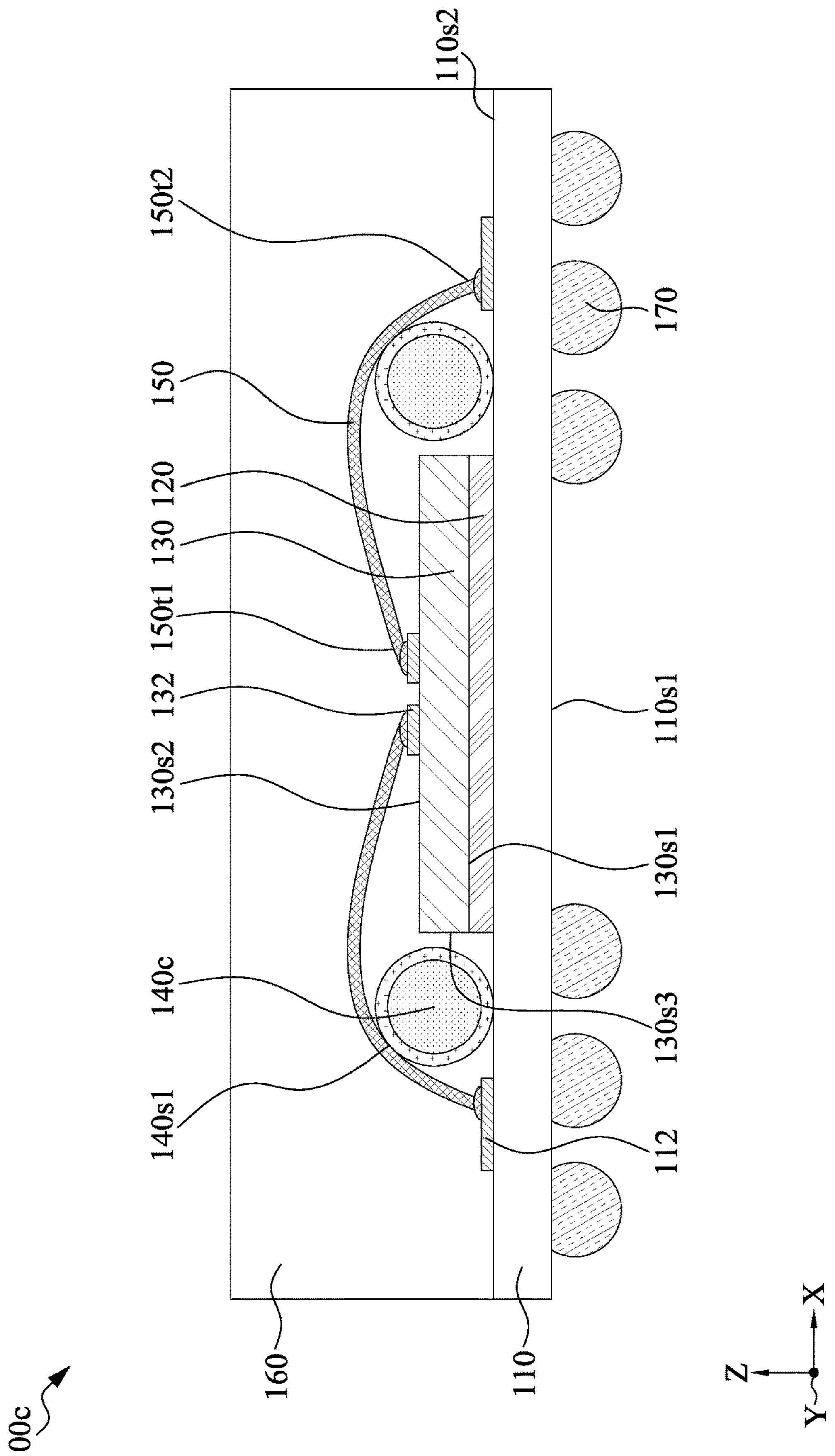

110s2
150t2
170
150
120
130
150t1
132
110s1
130s2
130s1
140c
130s3
140s1
112
160
110
100c
X
Z
Y

FIG. 3

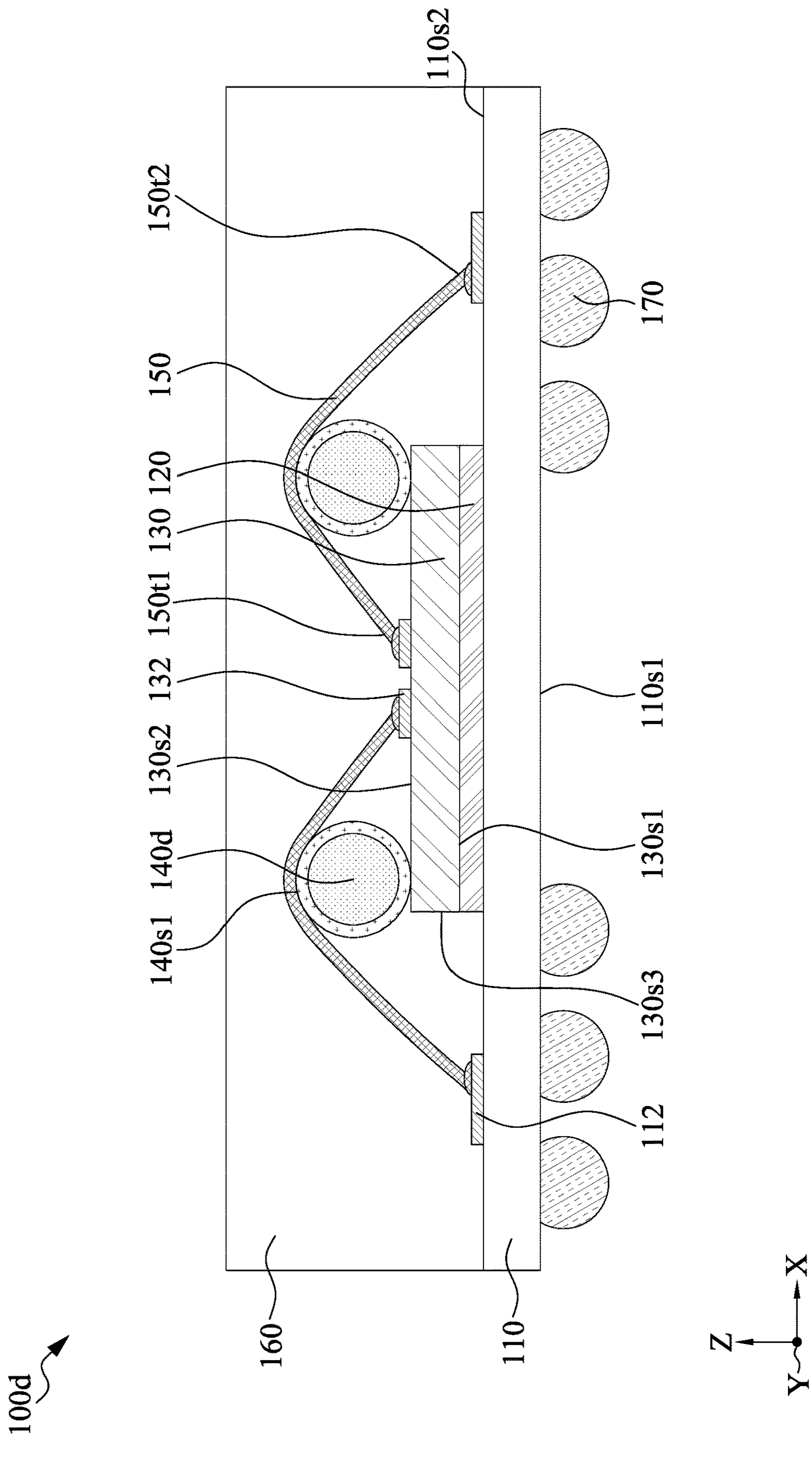

110s2
150t2
150
120
130
150t1
132
130s2
140d
140s1
170
110s1
130s1
130s3
112
160
110
100d
Z
X
Y

FIG. 4

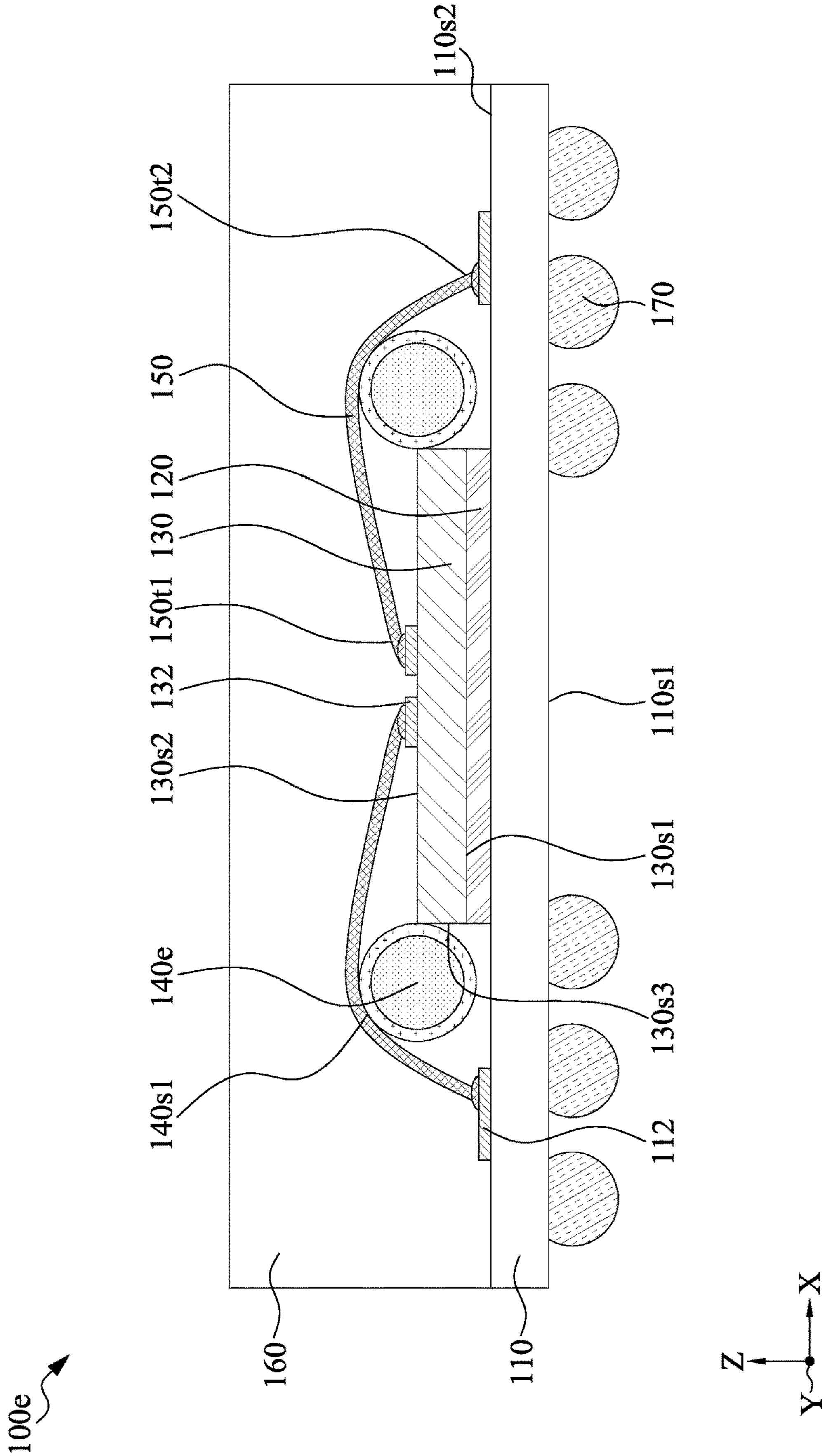

110s2
150t2
150
120
130
150t1
132
130s2
140e
140s1
170
110s1
130s1
130s3
112
160
110
100e
Z
X
Y

FIG. 5

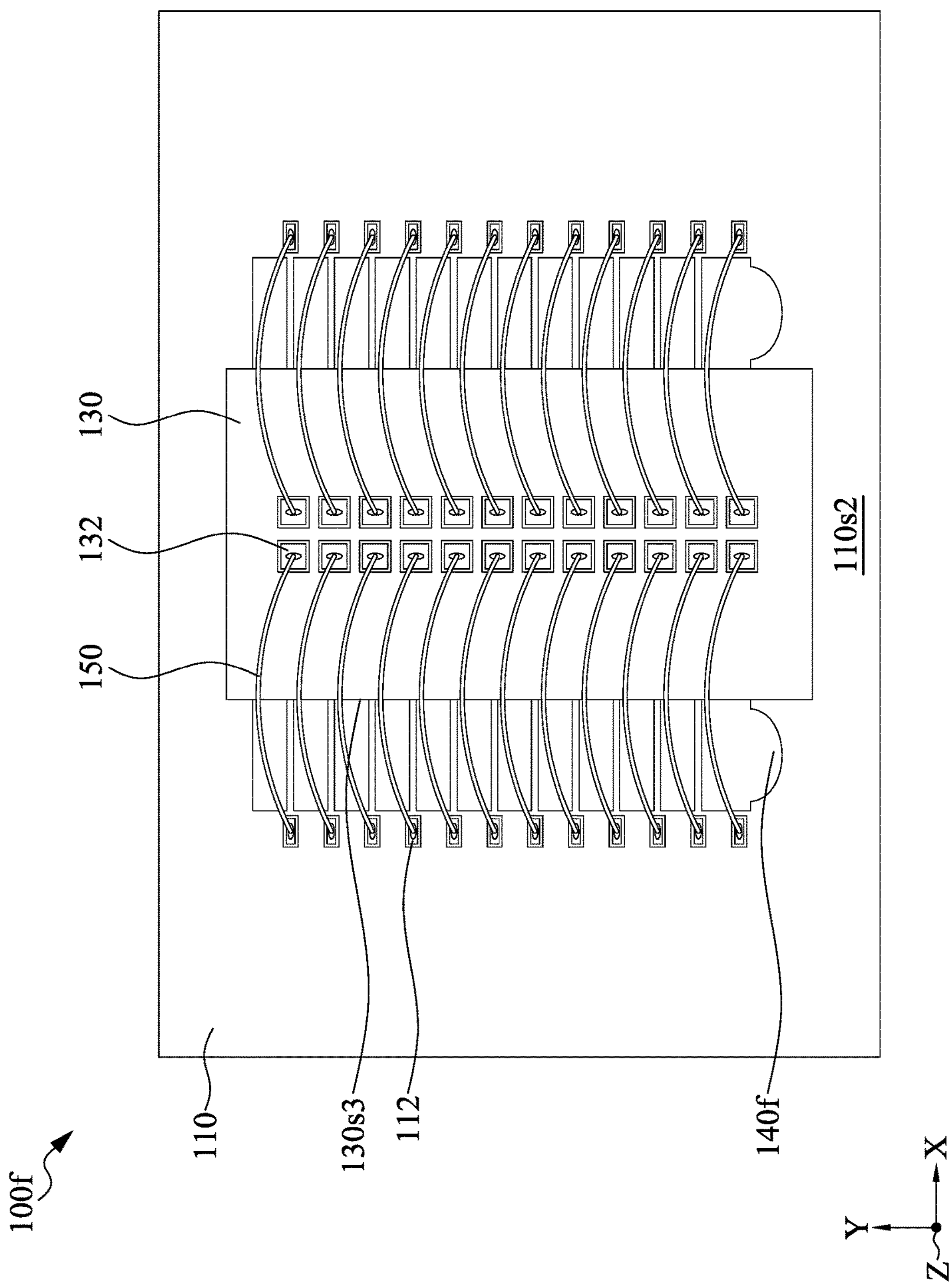



FIG. 6

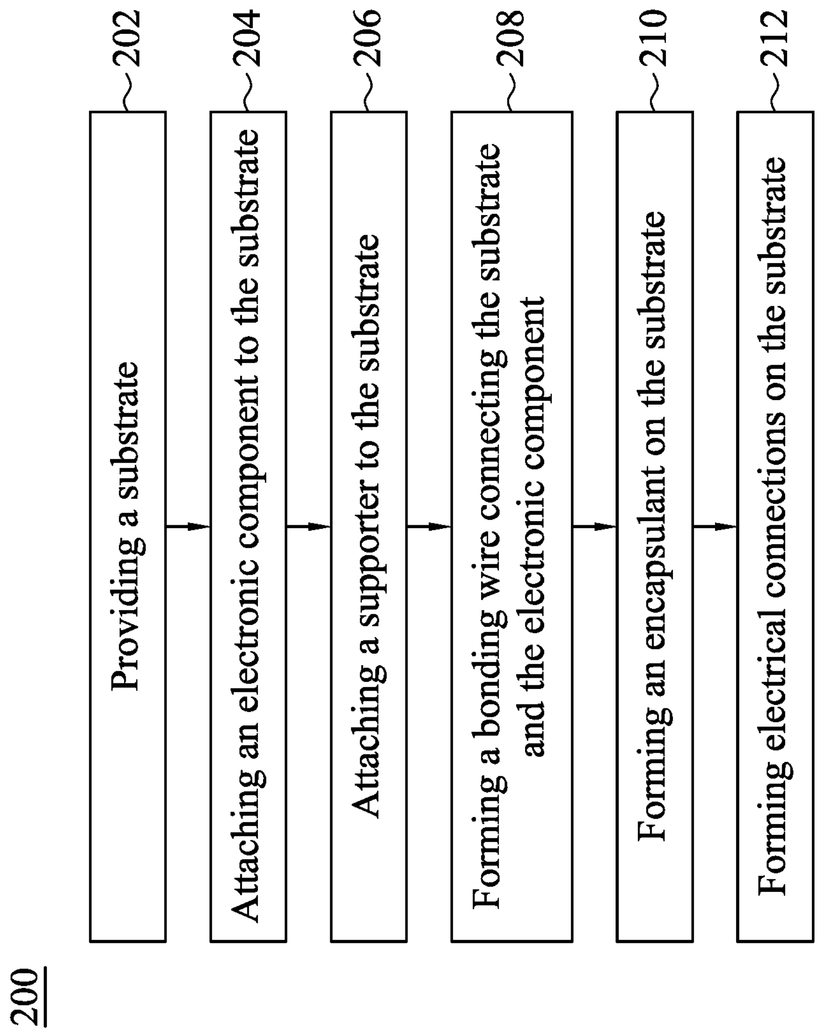

200
Providing a substrate
202
Attaching an electronic component to the substrate
204
Attaching a supporter to the substrate
206
Forming a bonding wire connecting the substrate and the electronic component
208
Forming an encapsulant on the substrate
210
Forming electrical connections on the substrate
212

FIG. 7

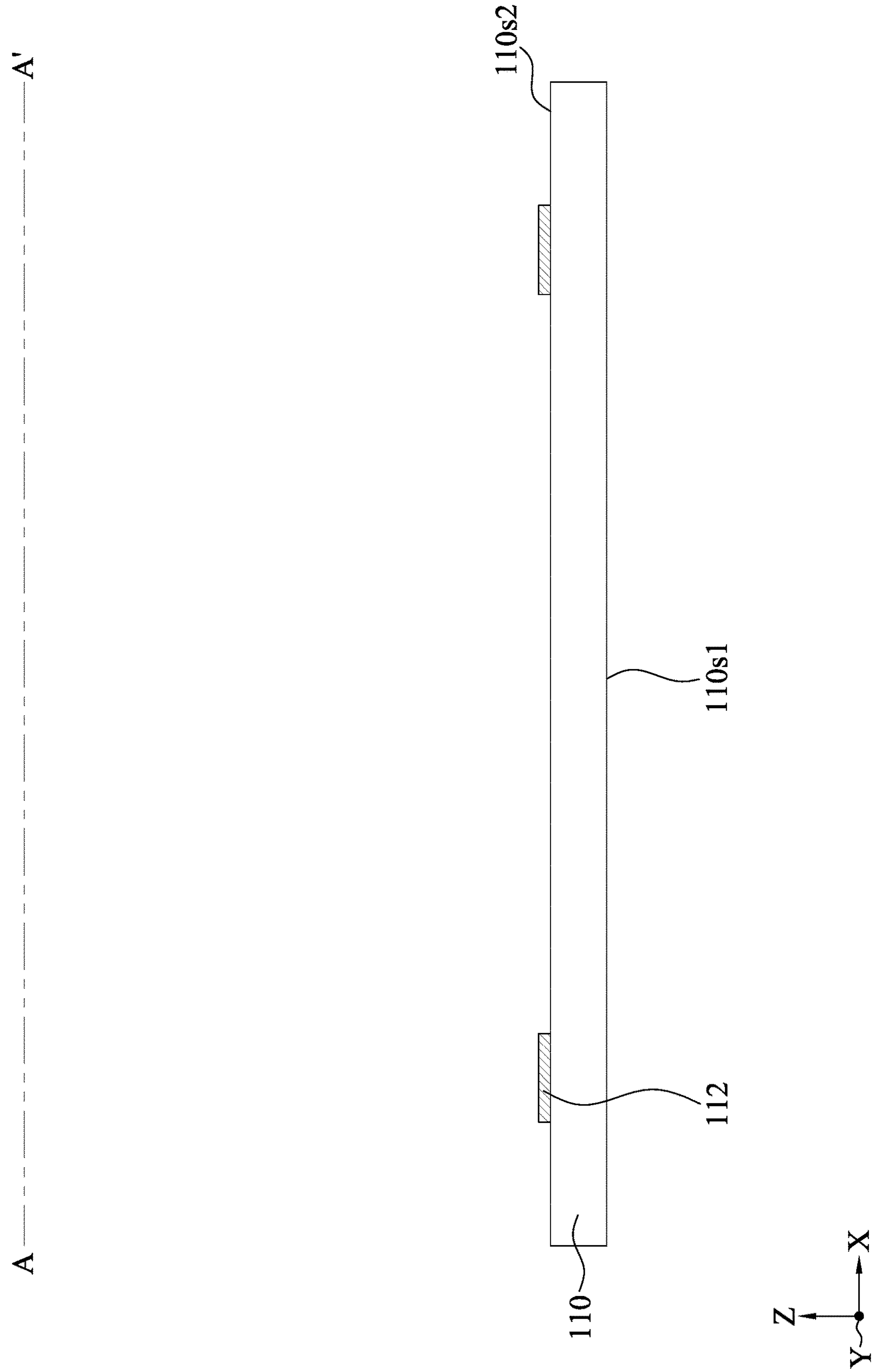



FIG. 8A

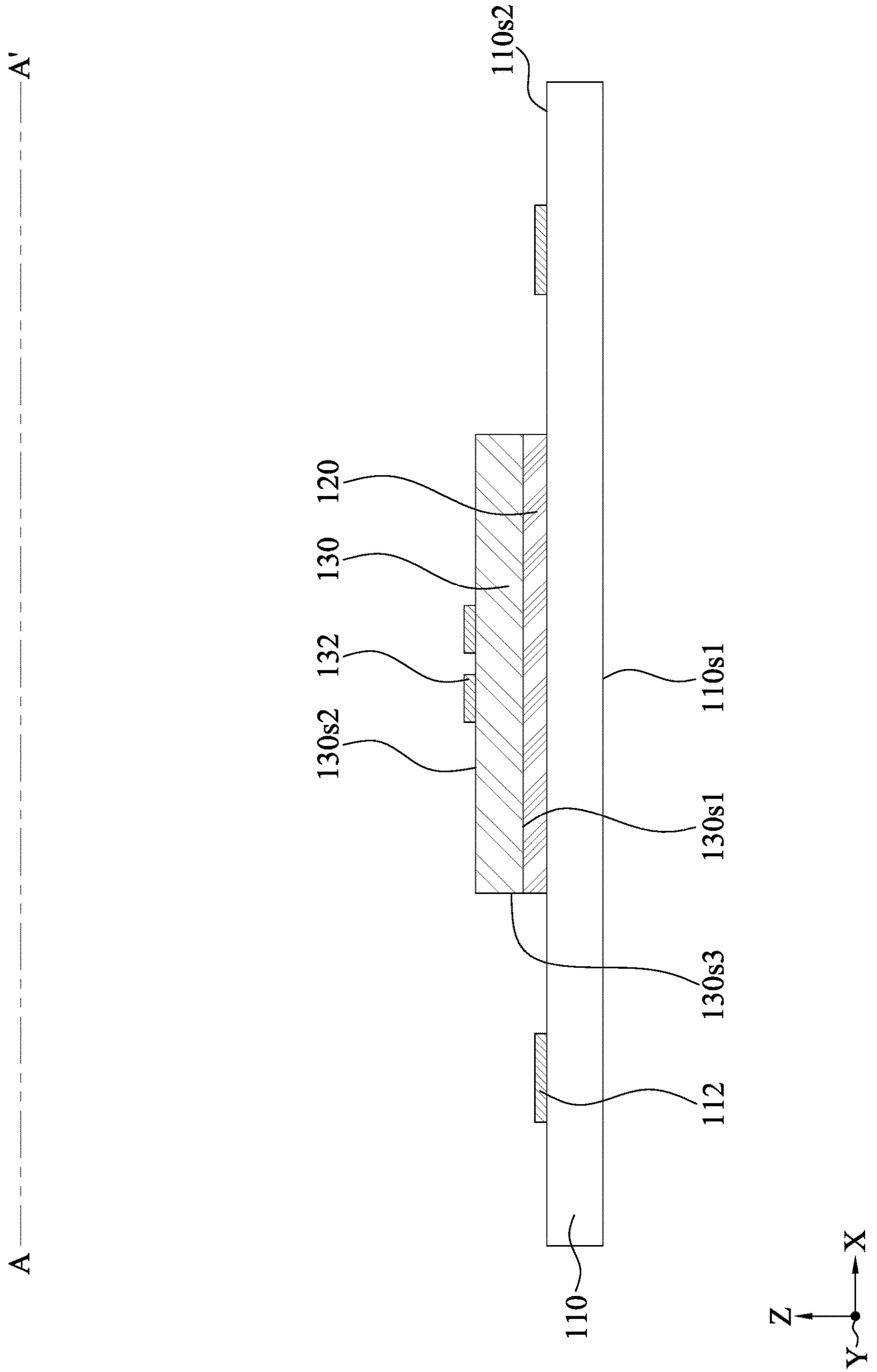
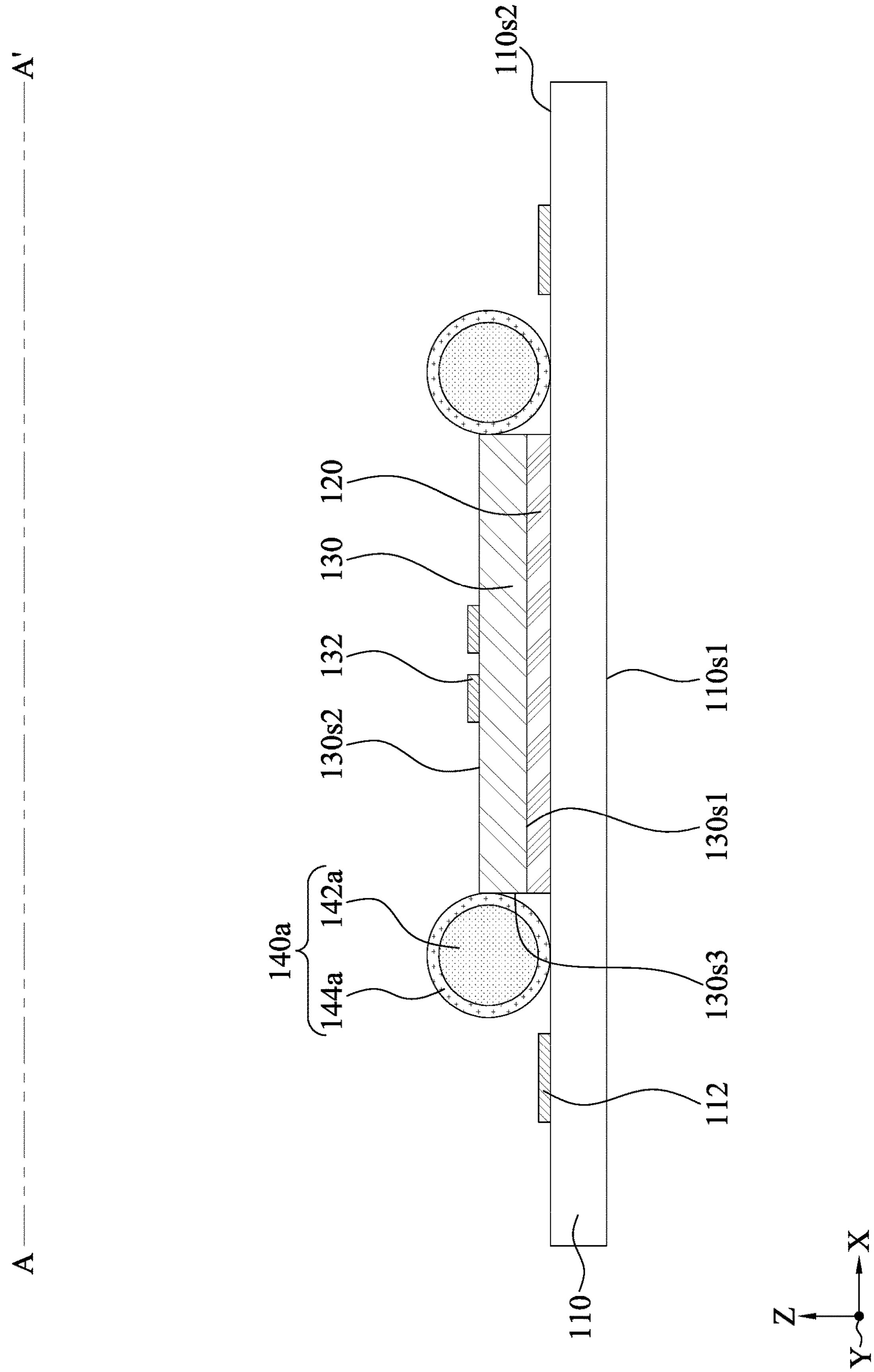



FIG. 8B



A
A'
110s2
120
130
132
130s2
110s1
130s1
140a
142a
144a
130s3
112
110
Z
X
Y


FIG. 8C

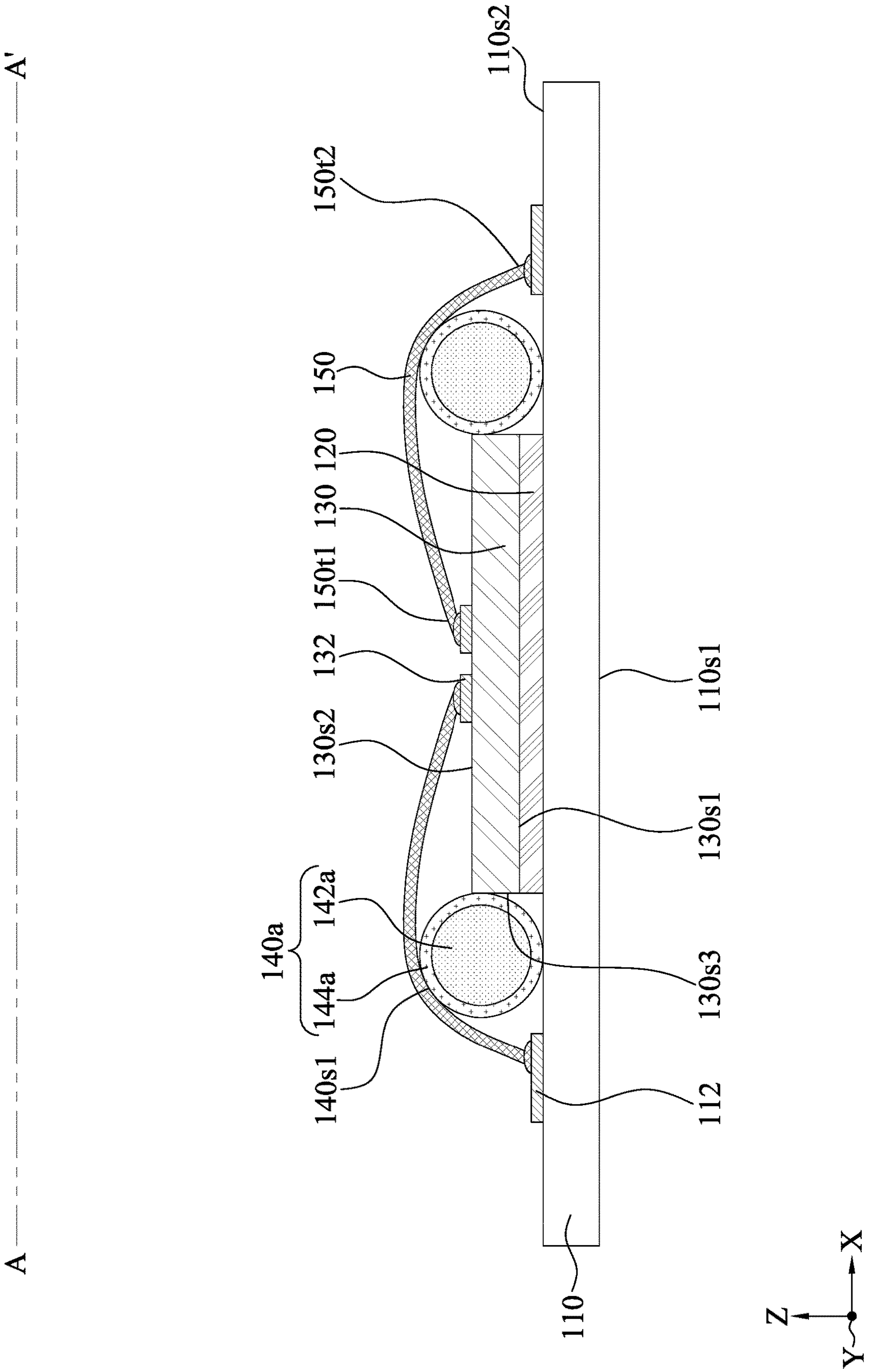

A
A'
110s2
150t2
150
120
130
150t1
132
130s2
142a
140a
144a
140s1
110s1
130s1
130s3
112
110
X
Z
Y

FIG. 8D

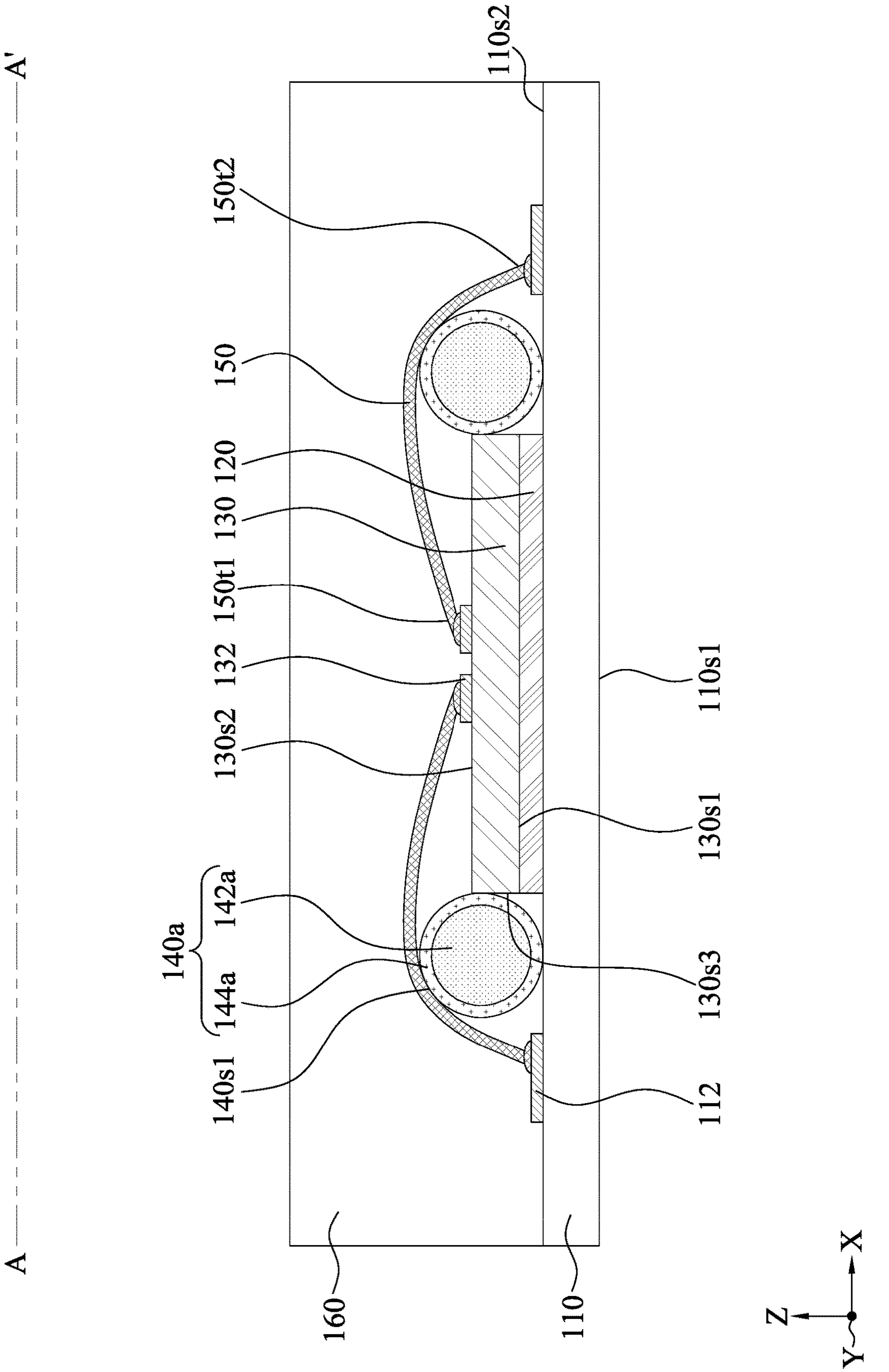

A'
A
110s2
150t2
150
120
130
150t1
132
130s2
142a
140a
144a
140s1
110s1
130s1
130s3
112
160
110
X
Z
Y

FIG. 8E

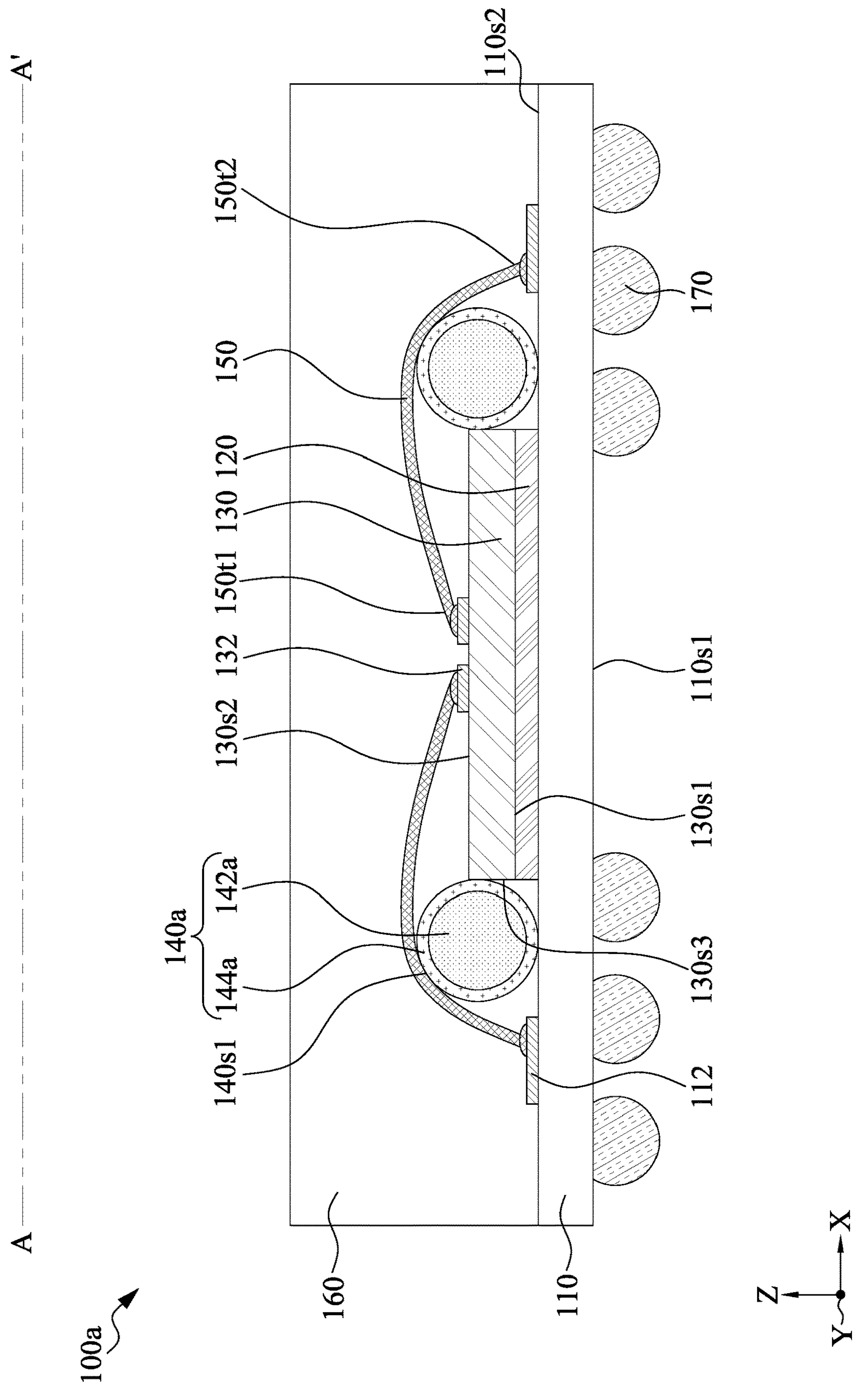

A'
A
110s2
150t2
170
150
120
130
150t1
132
130s2
110s1
130s1
142a
140a
144a
130s3
140s1
112
160
110
100a
X
Z
Y

FIG. 8F

# SEMICONDUCTOR DEVICE WITH SUPPORTER AGAINST WHICH BONDING WIRE IS DISPOSED AND METHOD FOR PREPARING THE SAME

## TECHNICAL FIELD

The present disclosure relates to a semiconductor device and method for manufacturing the same, and more particularly, to a semiconductor device including a supporter against which a bonding wire is disposed.

## DISCUSSION OF THE BACKGROUND

With the rapid growth of the electronics industry, integrated circuits (ICs) have achieved high performance and miniaturization. Technological advances in IC materials and design have produced generations of ICs in which each successive generation has smaller and more complex circuits.

Many techniques have been developed for integrating an electronic component and a substrate. For example, the electronic component and the substrate may be connected by a bonding wire. In order to prevent the bonding wire from being disposed against the corner of the electronic component, the bonding wire is lengthened, increasing the size of the semiconductor device and resistance thereof. Therefore, a new semiconductor device and method of improving such problems is required.

This Discussion of the Background section is provided for background information only. The statements in this Discussion of the Background are not an admission that the subject matter disclosed herein constitutes prior art with respect to the present disclosure, and no part of this Discussion of the Background may be used as an admission that any part of this application constitutes prior art with respect to the present disclosure.

## SUMMARY

One aspect of the present disclosure provides a semiconductor device. The semiconductor device includes a substrate, an electronic component, a bonding wire, and a supporter. The electronic component is disposed on the substrate. The bonding wire includes a first terminal connected to the electronic component and a second terminal connected to the substrate. The bonding wire is disposed against the supporter.

Another aspect of the present disclosure provides another semiconductor device. The semiconductor device includes a substrate, an electronic component, a bonding wire, and a supporter. The electronic component is disposed on the substrate. The bonding wire includes a first terminal connected to the electronic component and a second terminal connected to the substrate. The supporter is disposed between the first terminal and the second terminal of the bonding wire.

Another aspect of the present disclosure provides a method of manufacturing a semiconductor device. The method includes providing a substrate. The method also includes attaching an electronic component to the substrate. The method further includes attaching a supporter to the substrate. In addition, the method includes forming a bonding wire connecting the substrate and the electronic component. The electronic component is disposed against the bonding wire.



In embodiments of the present disclosure, the semiconductor device may include a supporter utilized to fix bonding wire(s). The supporter may be disposed between two terminals of the bonding wire. The supporter may provide a smooth area, such as a smooth surface, a smooth edge, or a smooth corner on which the bonding wire is disposed. Thus, the bonding wire may be disposed against the supporter with a tolerable tension. The length of the bonding wire may be reduced, resulting in a relatively small semiconductor device and relatively low resistance of the bonding wire.

The foregoing has outlined rather broadly the features and technical advantages of the present disclosure in order that the detailed description of the disclosure that follows may be better understood. Additional features and advantages of the disclosure will be described hereinafter, and form the subject of the claims of the disclosure. It should be appreciated by those skilled in the art that the conception and specific embodiment disclosed may be readily utilized as a basis for modifying or designing other structures or processes for carrying out the same purposes of the present disclosure. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the disclosure as set forth in the appended claims.

## BRIEF DESCRIPTION OF THE DRAWINGS

A more complete understanding of the present disclosure may be derived by referring to the detailed description and claims when considered in connection with the Figures, where like reference numbers refer to similar elements throughout the Figures, and:

FIG. **1**A is a top view of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **1**B is a cross-sectional view along line A-A' of the semiconductor device as shown in FIG. **1**A, in accordance with some embodiments of the present disclosure.

FIG. **2** is a cross-sectional view of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **3** is a cross-sectional view of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **4** is a cross-sectional view of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **5** is a cross-sectional view of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **6** is a top view of a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **7** is a flowchart illustrating a method of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

FIG. **8**A illustrates one or more stages of an exemplary method for manufacturing a semiconductor device according to some embodiments of the present disclosure.

FIG. **8**B illustrates one or more stages of an exemplary method for manufacturing a semiconductor device according to some embodiments of the present disclosure.

FIG. **8**C illustrates one or more stages of an exemplary method for manufacturing a semiconductor device according to some embodiments of the present disclosure.

FIG. **8**D illustrates one or more stages of an exemplary method for manufacturing a semiconductor device according to some embodiments of the present disclosure.

FIG. **8**E illustrates one or more stages of an exemplary method for manufacturing a semiconductor device according to some embodiments of the present disclosure.

FIG. **8**F illustrates one or more stages of an exemplary method for manufacturing a semiconductor device according to some embodiments of the present disclosure.

## DETAILED DESCRIPTION

Embodiments, or examples, of the disclosure illustrated in the drawings are now described using specific language. It shall be understood that no limitation of the scope of the disclosure is hereby intended. Any alteration or modification of the described embodiments, and any further applications of principles described in this document, are to be considered as normally occurring to one of ordinary skill in the art to which the disclosure relates. Reference numerals may be repeated throughout the embodiments, but this does not necessarily mean that feature(s) of one embodiment apply to another embodiment, even if they share the same reference numeral.

It shall be understood that, although the terms first, second, third, etc., may be used herein to describe various elements, components, regions, layers or sections, these elements, components, regions, layers or sections are not limited by these terms. Rather, these terms are merely used to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the present inventive concept.

The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limited to the present inventive concept. As used herein, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It shall be further understood that the terms "comprises" and "comprising," when used in this specification, point out the presence of stated features, integers, steps, operations, elements, or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or groups thereof.

FIG. **1**A and FIG. **1**B illustrate a semiconductor device **100***a*, in accordance with some embodiments of the present disclosure, wherein FIG. **1**A is a top view, and FIG. **1**B is a cross-sectional view along line A-A' of FIG. **1**A.

In some embodiments, the semiconductor device **100***a* may include a substrate **110**. In some embodiments, the substrate **110** may be or include, for example, a printed circuit board (PCB), such as a paper-based copper foil laminate, a composite copper foil laminate, or a polymer-impregnated glass-fiber-based copper foil laminate.

In some embodiments, the substrate **110** may include a surface **110***s***1** and a surface **110***s***2** opposite to the surface **110***s***1**. In some embodiments, the surface **110***s***1** may also be referred to as a lower surface. In some embodiments, the surface **110***s***2** may also be referred to as an upper surface.

In some embodiments, the substrate **110** may include conductive pad(s), trace(s), via(s), layer(s), or other interconnection(s). For example, the substrate **110** may include one or more transmission lines (e.g., communications cables) and one or more grounding lines and/or grounding planes. For example, the substrate **110** may include one or more conductive pads (e.g., **112**) in proximity to, adjacent to, or embedded in and exposed at the surface **110***s***1** and/or the surface **110***s***2** of the substrate **110**.

In some embodiments, the semiconductor device **100***a* may include an adhesive layer **120**. In some embodiments, the adhesive layer **120** may be disposed on the surface **110***s***2** of the substrate **110**.

In some embodiments, the semiconductor device **100***a* may include an electronic component **130**. In some embodiments, the electronic component **130** may be disposed on the surface **110***s***2** of the substrate **110**. In some embodiments, the electronic component **130** may be attached to the surface **110***s***2** of the substrate **110** through the adhesive layer **120**.

In some embodiments, the electronic component **130** may include a memory device, such as a dynamic random access memory (DRAM) device, a one-time programming (OTP) memory device, a static random access memory (SRAM) device, or other suitable memory devices. In some embodiments, the electronic component **130** may include a logic device (e.g., system-on-a-chip (SoC), central processing unit (CPU), graphics processing unit (GPU), application processor (AP), microcontroller, etc.), a radio frequency (RF) device, a sensor device, a micro-electro-mechanical-system (MEMS) device, a signal processing device (e.g., digital signal processing (DSP) device)), a front-end device (e.g., analog front-end (AFE) devices) or other devices.

The electronic component **130** may have a surface **130***s***1** and a surface **130***s***2** opposite to the surface **130***s***1**. In some embodiments, the surface **130***s***1** may also be referred to as a backside surface or a lower surface. In some embodiments, the surface **130***s***2** may also be referred to as an active surface or an upper surface. As used herein, the term "active surface" may refer to a surface on which terminal(s) is disposed for transmitting and/or receiving signals. In some embodiments, the surface **130***s***1** of the electronic component **130** may face the surface **110***s***2** of the substrate **110**. The electronic component **130** may have a surface **130***s***3** (or a lateral surface) extending between the surfaces **130***s***1** and **130***s***2** of the electronic component **130**.

In some embodiments, the electronic component **130** may include conductive pads **132**. The conductive pad **132** may be disposed on the surface **130***s***2** of the electronic component **130**. In some embodiments, the conductive pad **132** may include metal, such as copper (Cu), tungsten (W), silver (Ag), gold (Au), ruthenium (Ru), iridium (Ir), nickel (Ni), osmium (Os), ruthenium (Rh), aluminum (Al), molybdenum (Mo), cobalt (Co), alloys thereof, combinations thereof or other suitable materials.

In some embodiments, the semiconductor device **100***a* may include supporter(s) **140***a*. In some embodiments, the supporters **140***a* may be disposed on the surface **110***s***2** of the substrate **110**. In some embodiments, the supporters **140***a* may be disposed on two opposite surfaces **130***s***3** of the electronic component **130**. However, the present disclosure is not intended to be limiting.

In some embodiments, the supporter **140***a* may be configured to provide an area, such as a surface, an edge, or a corner, against which a bonding wire (or a conductive wire) is disposed. In some embodiments, at least a portion of the supporter **140***a* may have a rounded surface, a rounded edge, or a rounded corner against which a conductive wire is disposed. In some embodiments, at least a portion of the spacer **142***a* may have a smooth surface, a smooth edge, or a smooth corner against which a conductive wire is disposed. As used herein, the term "smooth surface, edge, or corner" may refer to a surface, edge, or corner which may be relatively blunt.

In some embodiments, the supporter **140**a may be in contact with the surface **110**s**2** of the substrate **110**. In some embodiments, the supporter **140**a may be in contact with the surface **130**s**3** of the electronic component **130**. In some embodiments, the supporter **140**a may be disposed against the electronic component **130**. In some embodiments, the supporter **140**a may be disposed against the surface **130**s**3** of the electronic component **130**. As used herein, the term "X is disposed against Y" may mean that X imposes a force or a stress, in addition to or other than a gravitational force, on Y.

In some embodiments, the supporter **140**a may include a spacer **142**a and an adhesive element **144**a. In some embodiments, at least a portion of the spacer **142**a may have a rounded surface, a rounded edge, or a rounded corner against which a conductive wire is disposed. In some embodiments, at least a portion of the spacer **142**a may have a smooth surface, a smooth edge, or a smooth corner against which a conductive wire is disposed. In some embodiments, the spacer **142**a may include an electrical insulative material or an electrically conductive material. In some embodiments, the spacer **142**a may include resin, glass, or other suitable materials. The spacer **142**a may include metal, alloy or other suitable materials. In some embodiments, the spacer **142**a may be circular, elliptical, or other profile shape.

In some embodiments, the adhesive element **144**a may cover an external surface (not annotated in the figures) of the spacer **142**a. In some embodiments, the adhesive element **144**a may enclose the spacer **142**a. In some embodiments, the adhesive element **144**a may be conformally disposed on the spacer **142**a. In some embodiments, the adhesive element **144**a may include an electrically insulative material.

In some embodiments, the semiconductor device **100**a may include bonding wire(s) **150**. In some embodiments, each of the bonding wires **150** may have a terminal **150**t**1** connected to (or bonded to) the surface **130**s**2** of the electronic component **130** and a terminal **150**t**2** connected to (or bonded to) the surface **110**s**2** of the substrate **110**. In some embodiments, the terminal **150**t**1** of the bonding wire **150** may be bonded to the conductive pad **132** of the electronic component **130**. In some embodiments, the terminal **150**t**2** of the bonding wire **150** may be bonded to the conductive pad **112** of the substrate **110**. In some embodiments, the supporter **140**a may be disposed between the terminals **150**t**1** and **150**t**2** of the bonding wire **150**. In some embodiments, the bonding wire **150** may include metal, such as copper (Cu), silver (Ag), gold (Au), nickel (Ni), aluminum (Al), alloys thereof, combinations thereof, or other suitable materials.

In some embodiments, the bonding wire **150** may be disposed on the supporter **140**a. In some embodiments, the bonding wire **150** may be disposed against the supporter **140**a. In some embodiments, the bonding wire **150** may be disposed against a smooth (or rounded) area **140**s**1**, such as surface, edge, or corner, of the supporter **140**a. In some embodiments, the bonding wire **150** may be in contact with the supporter **140**a, resulting in force or a stress imposed on the supporter **140**a. Thus, the supporter **140**a may be disposed against the electronic component **130** through the supporter **140**a.

The terminal **150**t**1** of the bonding wire **150** and the surface **130**s**3** of the electronic component **130** may have a length L**1**, along the X-axis, therebetween. The terminal **150**t**2** of the bonding wire **150** and the surface **130**s**3** of the electronic component **130** may have a length L**2**, along the X-axis, therebetween. In some embodiments, the length L**1** may substantially equal or exceed the length L**2**. In some embodiments, a ratio between the length L**1** and the length L**2** may range from about 1.1 to about 3, such as 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, or 3.

When the ratio between the lengths L**1** and L**2** ranges from about 1.1 to about 3, the bonding wire **150** may be disposed against the supporter **140**a with relatively little tension, which thereby prevents the bonding wire **150** from being broken. Further, the supporter **140**a may result in a relatively small sum of the lengths L**1** and L**2**, which thereby reduces the size of the semiconductor device **100**a. Further, as the length of the bonding wire **150** decreases, the resistance of the bonding wire is correspondingly reduced.

The surface **130**s**2** of the electronic component **130** and the terminal **150**t**1** (or terminal **150**t**2**) of the bonding wire **150** may have a length L**3**, along the Y-axis, therebetween. The supporter **140**a may have a length L**4** (or thickness or diameter) along the Y-axis. In some embodiments, the length L**4** may exceed the length L**3**. In some embodiments, the ratio between the lengths L**3** and the L**4** may range from about 1.2 to about 1.8, such as 1.2, 1.3, 1.4, 1.5, 1.6, 1.7 or 1.8.

When the ratio between the lengths L**3** and L**4** ranges from about 1.2 to about 1.8, the bonding wire **150** may be disposed against the supporter **140**a with relatively little tension, which thereby prevents the bonding wire **150** from being broken.

As shown in FIG. **1**A, multiple bonding wires **150** may share a common supporter **140**a. In some embodiments, the supporter **140**a may be in contact with a plurality of bonding wires **150**. In some embodiments, the bonding wire **150** may extend along the X-axis. In some embodiments, the supporter **140**a may extend along the Y-axis.

In some embodiments, the semiconductor device **100**a may include an encapsulant **160**. In some embodiments, the encapsulant **160** may be disposed on the surface **110**s**2** of the substrate **110**. In some embodiments, the encapsulant **160** may cover the surface **110**s**2** of the substrate **110**. In some embodiments, the encapsulant **160** may encapsulate the supporter **140**a. In some embodiments, the encapsulant **160** may encapsulate the bonding wire **150**. The encapsulant **160** may include insulative or dielectric material. In some embodiments, the encapsulant **160** may be made of molding material that may include, for example, a Novolac-based resin, an epoxy-based resin, a silicone-based resin, or other suitable encapsulant. Suitable fillers may also be included, such as powdered $SiO_2$.

In some embodiments, the semiconductor device **100**a may include electrical connections **170**. The electrical connection **170** may be disposed on the surface **110**s**1** of the substrate **110**. In some embodiments, the electrical connection **170** may be configured to electrically connect the semiconductor device **100**a and an external device (not shown). In some embodiments, the electrical connection **170** may include solder material, such as alloys of gold and tin solder or alloys of silver and tin solder.

In a comparative example, bonding wires are connected between an electronic component and a substrate without a supporter. In such condition, the length of the bonding wire should be greater than a predetermined value in order to prevent the bonding wire from being disposed against the corner of the electronic component. If not, the bonding wire may be prone to breakage due to relatively high tension. Therefore, the comparative example may have a relatively large width. In embodiments of the present disclosure, the bonding wires can be disposed against a smooth area, such as a surface, an edge, or a corner, of the supporter with relatively little tension. Further, the length of the bonding wire may be reduced, resulting in a relatively small size of the semiconductor device and a relatively small resistance.

FIG. **2** is a cross-sectional view of a semiconductor device **100***b*, in accordance with some embodiments of the present disclosure. The semiconductor device **100***b* is similar to the semiconductor device **100***a* as shown in FIG. **1**B, and the differences therebetween are described below.

In some embodiments, the semiconductor device **100***b* may include a supporter **140***b*. The supporter **140***b* may include a spacer **142***b*. In some embodiments, the spacer **142***b* may be rectangular, square, or other suitable profile shape. In some embodiments, the supporter **140***b* may have a rounded surface, a rounded edge, or a rounded corner against which the bonding wire **150** is disposed. In some embodiments, at least a portion of the supporter **140***b* may have a smooth surface, a smooth edge, or a smooth corner against which the bonding wire **150** is disposed.

In some embodiments, the supporter **140***b* may be fully in contact with the surface **130***s***3** of the electronic component **130**. In some embodiments, at least a portion of the spacer **142***b* may have a rounded surface, a rounded edge, or a rounded corner against which the bonding wire **150** is disposed. In some embodiments, at least a portion of the spacer **142***b* may have a smooth surface, a smooth edge, or a smooth corner against which the bonding wire **150** is disposed. Therefore, the bonding wire **150** may have relatively little tension imposed on the bonding wire **150**.

FIG. **3** is a cross-sectional view of a semiconductor device **100***c*, in accordance with some embodiments of the present disclosure. The semiconductor device **100***c* is similar to the semiconductor device **100***a* as shown in FIG. **1**B, and the differences therebetween are described below.

In some embodiments, the semiconductor device **100***c* may include a supporter **140***c*. In some embodiments, the supporter **140***c* may be disposed on the surface **110***s***2** of the substrate **110** and spaced apart from the surface **130***s***3** of the electronic component **130**. In some embodiments, the supporter **140***c* may provide a smooth (or rounded) surface, edge, or corner against which the bonding wire **150** is disposed. In some embodiments, at least a portion of the supporter **140***c* may have a smooth surface, a smooth edge, or a smooth corner against which the bonding wire **150** is disposed.

When the supporter **140***c* is spaced apart from the electronic component **130**, the thickness (or diameter) of the supporter **140***c* may be reduced. Therefore, the size of the semiconductor device **100***c* may be reduced.

FIG. **4** is a cross-sectional view of a semiconductor device **100***d*, in accordance with some embodiments of the present disclosure. The semiconductor device **100***d* is similar to the semiconductor device **100***a* as shown in FIG. **1**B, and the differences therebetween are described below.

In some embodiments, the semiconductor device **100***d* may include a supporter **140***d*. In some embodiments, the supporter **140***d* may be disposed on the surface **130***s***2** of the electronic component **130**. In some embodiments, the supporter **140***d* may be disposed against the surface **130***s***2** of the electronic component **130**. In some embodiments, the supporter **140***d* may be in contact with the surface **130***s***2** of the electronic component **130**. In some embodiments, the supporter **140***c* may be spaced apart from the surface **110***s***2** of the substrate **110**. In some embodiments, the supporter **140***d* may provide a smooth (or rounded) surface, edge, or corner against which the bonding wire **150** is disposed.

When the supporter **140***d* is disposed on the electronic component **130**, the length of the bonding wire **150** along the X-axis may be reduced. Therefore, the size of the semiconductor device **100***d* may be reduced.

FIG. **5** is a cross-sectional view of a semiconductor device **100***e*, in accordance with some embodiments of the present disclosure. The semiconductor device **100***e* is similar to the semiconductor device **100***a* as shown in FIG. **1**B, and the differences therebetween are described below.

In some embodiments, the semiconductor device **100***e* may include a supporter **140***e*. In some embodiments, the supporter **140***e* may be disposed on the surface **130***s***3** of the electronic component **130**. In some embodiments, the supporter **140***e* may be in contact with the surface **130***s***3** of the substrate **110**. In some embodiments, the supporter **140***e* may be spaced apart from the surface **130***s***2** of the electronic component **130**. In some embodiments, the supporter **140***e* may be spaced apart from the surface **110***s***2** of the substrate **110**. In some embodiments, the supporter **140***e* may provide a smooth (or rounded) surface, edge, or corner against which the bonding wire **150** is disposed.

The tension of the bonding wire **150** may be adjusted by modifying the location of the supporter **140***e*. Thus, the length and the tension of the bonding wire **150** may be optimized.

FIG. **6** is a top view of a semiconductor device **100***f*, in accordance with some embodiments of the present disclosure. The semiconductor device **100***f* is similar to the semiconductor device **100***a* as shown in FIG. **1**A, and the differences therebetween are described below.

In some embodiments, the semiconductor device **100***f* may include a plurality of supporters **140***f*. In some embodiments, each of the supporters **140***f* may provide a smooth (or rounded) surface, edge, or corner against which the bonding wire **150** is disposed. Each of the supporters **140***f* may correspond to one of the bonding wires **150**. In some embodiments, each of the supporters **140***f* may be in contact with a corresponding bonding wire **150**.

FIG. **7** is a flowchart illustrating a method **200** of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

The method **200** begins with operation **202** in which a substrate may be provided. The substrate may have a lower surface and an upper surface opposite to the lower surface. The substrate may include one or more conductive pads in proximity to, adjacent to, or embedded in and exposed at the lower surface and/or the upper surface of the substrate.

The method **200** continues with operation **204** in which an electronic component may be formed on the upper surface of the substrate. In some embodiments, the electronic component may be attached to the upper surface of the substrate by an adhesive layer. The electronic component may have a rear surface, an active surface, and a lateral surface extending between the rear surface and active surface of the electronic component. The electronic component may have a conductive pad on the active surface of the electronic component.

The method **200** continues with operation **206** in which a supporter may be formed on the upper surface of the substrate. In some embodiments, the supporter may be in contact with the upper surface of the substrate. In some embodiments, the supporter may be in contact with the lateral surface of the electronic component. In some embodiments, at least a portion of the supporter may have a rounded surface, a rounded edge, or a rounded corner. In some embodiments, at least a portion of the supporter may have a smooth surface, a smooth edge, or a smooth corner.

In some embodiments, the supporter may have a spacer and an adhesive element. In some embodiments, at least a portion of the spacer may have a rounded surface, a rounded edge, or a rounded corner. In some embodiments, at least a portion of the spacer may have a smooth surface, a smooth edge, or a smooth corner. In some embodiments, the adhesive element is conformally formed on the spacer.

The method **200** continues with operation **208** in which a bonding wire may be formed. In some embodiments, the bonding wire may have a first terminal connected to the electronic component and a second terminal connected to the substrate. In some embodiments, the bonding wire may be disposed against the supporter. In some embodiments, a portion of the bonding wire may be in contact with the supporter. In some embodiments, at least a portion of the bonding wire may be disposed against a smooth area, such as a surface, an edge, or a corner, of the supporter.

The method **200** continues with operation **210** in which an encapsulant may be formed on the upper surface of the substrate. In some embodiments, the encapsulant may encapsulate the electronic component, the supporter, and the bonding wire.

The method **200** continues with operation **212** in which electrical connections may be formed on the lower surface of the substrate, which thereby produces a semiconductor device.

In embodiments of the present disclosure, the semiconductor device may include a supporter. The supporter may be disposed between two terminals of the bonding wire. The bonding wire may be disposed against a smooth area, such as a surface, an edge, or a corner, of the supporter with relatively little tension. Further, the length of the bonding wire may be reduced, resulting in a relatively small size of the semiconductor device and a relatively small resistance of the bonding wire.

The method **200** is merely an example, and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, or after each operation of the method **200**, and some operations described can be replaced, eliminated, or reordered for additional embodiments of the method. In some embodiments, the method **200** can include further operations not depicted in FIG. **7**. In some embodiments, the method **200** can include one or more operations depicted in FIG. **7**.

FIG. **8**A-FIG. **8**F illustrate one or more stages of an exemplary method for manufacturing a semiconductor device according to some embodiments of the present disclosure. In some embodiments, the semiconductor device **100***a* may be manufactured through the operations described with respect to FIG. **8**A-FIG. **8**F.

Referring to FIG. **8**A, a substrate **110** may be provided. The substrate **110** may have a surface **110***s***1** and a surface **110***s***2** opposite to the surface **110***s***1**. The substrate **110** may include one or more conductive pads (e.g., **112**) in proximity to, adjacent to, or embedded in and exposed at the surface **110***s***1** and/or the surface **110***s***2** of the substrate.

Referring to FIG. **8**B, an electronic component **130** may be formed on the surface **110***s***2** of the substrate **110**. In some embodiments, the electronic component **130** may be attached to the surface **110***s***2** of the substrate **110** by an adhesive layer **120**. The electronic component **130** may have a surface **130***s***1**, a surface **130***s***2**, and a surface **130***s***3** extending between the surfaces **130***s***1** and **130***s***2** of the electronic component **130**. The electronic component **130** may have a conductive pad **132** on the surface **130***s***2** of the electronic component **130**.

Referring to FIG. **8**C, a supporter **140***a* may be formed on the surface **110***s***2** of the substrate **110**. In some embodiments, the supporter **140***a* may be in contact with the surface **110***s***2** of the substrate **110**. In some embodiments, the supporter **140***a* may be in contact with the surface **130***s***3** of the electronic component **130**. In some embodiments, at least a portion of the supporter **140***a* may have a rounded surface, a rounded edge, or a rounded corner. In some embodiments, at least a portion of the supporter **140***a* may have a smooth surface, a rounded edge, or a rounded corner.

In some embodiments, the supporter **140***a* may have a spacer **142***a* and an adhesive element **144***a*. In some embodiments, at least a portion of the spacer **142***a* may have a rounded surface, a rounded edge, or a rounded corner. In some embodiments, at least a portion of the spacer **142***a* may have a smooth surface, a smooth edge, or a smooth corner. In some embodiments, the adhesive element **144***a* is conformally formed on the spacer **142***a*.

Referring to FIG. **8**D, a bonding wire **150** may be formed. In some embodiments, the bonding wire **150** may have a terminal **150***t***1** connected to the electronic component **130** and a terminal **150***t***2** connected to the substrate **110**. In some embodiments, the bonding wire **150** may be disposed against the supporter **140***a*. In some embodiments, at least a portion of the bonding wire **150** may be disposed against a smooth area **140***s***1**, such as a surface, an edge, or a corner, of the supporter **140***a*. In some embodiments, a portion of the bonding wire **150** may be in contact with the supporter **140***a*.

Referring to FIG. **8**E, an encapsulant **160** may be formed on the surface **110***s***2** of the substrate **110**. The encapsulant **160** may be formed by a molding operation. In some embodiments, the encapsulant **160** may encapsulate the electronic component **130**, the supporter **140***a*, and the bonding wire **150**.

Referring to FIG. **8**F, electrical connections **170** may be formed on the surface **110***s***1** of the substrate **110**, which thereby produces the semiconductor device **100***a*. The electrical connection **170** may include a solder material.

It is contemplated that in FIG. **8**C, if the supporter is disposed on the surface **110***s***2** of the substrate **110** and spaced apart from the surface **130***s***3** of the electronic component **130**, a semiconductor device same or similar to the semiconductor device **100***c* as illustrated and described with reference to FIG. **3** can be formed.

It is contemplated that in FIG. **8**C, if the supporter is disposed on the surface **130***s***2** of the electronic component **130** and spaced apart from the substrate **110**, a semiconductor device same or similar to the semiconductor device **100***d* as illustrated and described with reference to FIG. **4** can be formed.

It is contemplated that in FIG. **8**C, if the supporter is disposed on the surface **130***s***3** of the electronic component **130** and spaced apart from the substrate **110**, a semiconductor device same or similar to the semiconductor device **100***e* as illustrated and described with reference to FIG. **5** can be formed.

It is contemplated that in FIG. **8**C, if a plurality of supporters, each of which may be utilized to fix a corresponding bonding wire **150**, are disposed on the surface **110***s***2** of the substrate **110**, a semiconductor device the same as or similar to the semiconductor device **100***f* as illustrated and described with reference to FIG. **6** can be formed.

In embodiments of the present disclosure, the semiconductor device may include a supporter. The supporter may be disposed between two terminals of the bonding wire. The bonding wire may be disposed against a smooth area, such as a surface, an edge, or a corner, of the supporter with relatively little tension. Further, the length of the bonding wire may be reduced, resulting in a relatively small size of the semiconductor device and a relatively small resistance of the bonding wire.

One aspect of the present disclosure provides a semiconductor device. The semiconductor device includes a substrate, an electronic component, a bonding wire, and a supporter. The electronic component is disposed on the substrate. The bonding wire includes a first terminal connected to the electronic component and a second terminal connected to the substrate. The bonding wire is disposed against the supporter.

Another aspect of the present disclosure provides another semiconductor device. The semiconductor device includes a substrate, an electronic component, a bonding wire, and a supporter. The electronic component is disposed on the substrate. The bonding wire includes a first terminal connected to the electronic component and a second terminal connected to the substrate. The supporter is disposed between the first terminal and the second terminal of the bonding wire.

Another aspect of the present disclosure provides a method of manufacturing a semiconductor device. The method includes providing a substrate. The method also includes attaching an electronic component to the substrate. The method further includes attaching a supporter to the substrate. In addition, the method includes forming a bonding wire connecting the substrate and the electronic component. The electronic component is disposed against the bonding wire.

In embodiments of the present disclosure, the semiconductor device may include a supporter. The supporter may be disposed between two terminals of the bonding wire. The bonding wire may be disposed against a smooth area, such as a surface, an edge, or a corner, of the supporter with relatively little tension. Further, the length of the bonding wire may be reduced, resulting in a relatively small size of the semiconductor device and a relatively small resistance of the bonding wire.

Although the present disclosure and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, many of the processes discussed above can be implemented in different methodologies and replaced by other processes, or a combination thereof.

Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

What is claimed is:

**1**. A semiconductor device, comprising:

a substrate;

an electronic component disposed on a top surface of the substrate;

a bonding wire comprising a first terminal connected to a top surface of the electronic component and a second terminal connected to the top surface of the substrate; and

a supporter having a circular cross-sectional shape and defining a bottom point in contact with the top surface of the substrate, and positioning adjacent to a lateral surface of the electronic component, wherein the bonding wire is disposed against the supporter, wherein a diameter of the supporter is greater than a thickness of the electronic component, such that the supporter has a contacting surface at a highest point above the electronic component for the bonding wire contacting with the supporter in a tolerable tension manner so as to minimize a length of the bonding wire between the first terminal and the second terminal, wherein the contacting surface of the supporter is located out of the electronic component, the bonding wire is extended and biased against the contacting surface of the supporter in a tolerable tension manner, and a peak of the bonding wire is in contact with the contacting surface of the supporter away from the electronic component;

wherein the bonding wire has a first portion extended from the first terminal to contact with the contacting surface of the supporter and a second portion extended from the contacting surface of the supporter to the second terminal.

**2**. The semiconductor device of claim **1**, wherein the supporter and the electronic component are arranged side-by-side to ensure the peak of the bonding wire away from the electronic component.

**3**. The semiconductor device of claim **1**, wherein the supporter is in contact with the electronic component.

**4**. The semiconductor device of claim **3**, wherein the contacting surface of the supporter is a round surface defined as an arc surface of the supporter.

**5**. The semiconductor device of claim **3**, wherein the supporter is in contact with the lateral surface of the electronic component only to keep the top surface of the electronic component free from the supporter.

**6**. The semiconductor device of claim **5**, wherein the supporter is partially in contact with the lateral surface of the electronic component.

**7**. The semiconductor device of claim **5**, wherein the supporter has a lateral point in contact with the lateral surface of the electronic component.

**8**. The semiconductor device of claim **3**, wherein the supporter is disposed against the lateral surface of the electronic component.

**9**. The semiconductor device of claim **1**, wherein the supporter is spaced apart from the lateral surface of the electronic component.

**10**. The semiconductor device of claim **1**, further comprising an encapsulant disposed on the top surface of the substrate to encapsulate the supporter, the bonding wire and the electronic component.

**11**. The semiconductor device of claim **1**, wherein the supporter comprises a spacer having a circular cross-sectional shape, and an adhesive element covering an external circumferential surface of the spacer, such that the spacer is enclosed within the adhesive element.

**12**. The semiconductor device of claim **1**, wherein the first terminal of the bonding wire and the lateral surface of the electronic component has a first length along a first direction substantially perpendicular to the lateral surface of the electronic component, and the second terminal and the lateral surface of the electronic component has a second length, along the first direction, less than the first length, wherein the peak of the bonding wire is supported between the second terminal of the bonding wire and the lateral surface of the electronic component to support the bonding wire in a tension manner.

**13**. The semiconductor device of claim **12**, wherein a ratio between the first length and the second length ranges from about 1.1 to about 3.

**14**. The semiconductor device of claim **1**, wherein an upper surface of the electronic component and the second terminal of the bonding wire has a third distance along a second direction substantially perpendicular to the upper surface of the electronic component, the supporter has the diameter along the second direction, and the diameter of the supporter is greater than the third distance.

**15**. The semiconductor device of claim **1**, wherein the diameter of the supporter is 1.5 times greater than the first distance.

**16**. A method of manufacturing a semiconductor device, comprising:

providing a substrate;

attaching an electronic component onto a top surface of the substrate;

attaching a supporter to the substrate adjacent to the electronic component, wherein the supporter has a circular cross-sectional shape and defines a bottom point in contact with the top surface of the substrate, wherein the supporter is positioned adjacent to a lateral surface of the electronic component; and

forming a bonding wire to have a first terminal connecting to a top surface of the electronic component and a second terminal connecting the top surface of the substrate,

wherein the bonding wire is disposed against the supporter, wherein a diameter of the supporter is greater than a thickness of the electronic component, such that the supporter has a contacting surface at a highest point above the electronic component for the bonding wire contacting with the supporter, wherein the contacting surface of the supporter is located out of the electronic component, the bonding wire runs across the supporter, and a peak of the bonding wire is in contact with the contacting surface of the supporter away from the electronic component;

wherein the bonding wire is extended and biased against the contacting surface of the supporter by:

extending a first portion of the bonding wire from the first terminal to contact with the contacting surface of the supporter; and

extending a second portion of the bonding wire from the contacting surface of the supporter to the second terminal, such that the bonding wire is extended in a tolerable tension manner to minimize a length of the bonding wire between the first terminal and the second terminal.

**17**. The method of claim **16**, wherein the supporter is spaced apart from the lateral surface of the electronic component.

**18**. The method of claim **16**, wherein the supporter has a lateral point in contact with the lateral surface of the electronic component.

**19**. The method of claim **16**, wherein the supporter is attached to the top surface of the substrate and the lateral surface of the electronic component by an adhesive element.

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